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A lightweight deep learning approach for detecting electrocardiographic lead misplacement

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Abstract

Objective. Electrocardiographic (ECG) lead misplacement can result in distorted waveforms and amplitudes, significantly impacting accurate interpretation. Although lead misplacement is a relatively low-probability event, with an incidence ranging from 0.4% to 4%, the large number of ECG records in clinical practice necessitates the development of an effective detection method. This paper aimed to address this gap by presenting a novel lead misplacement detection method based on deep learning models. *Approach*. We developed two novel lightweight deep learning model for limb and chest lead misplacement detection, respectively. For limb lead misplacement detection, two limb leads and V6 were used as inputs, while for chest lead misplacement detection, six chest leads were used as inputs. Our models were trained and validated using the Chapman database, with an 8:2 train-validation split, and evaluated on the PTB-XL, PTB, and LUDB databases. Additionally, we examined the model interpretability on the LUDB databases. Limb lead misplacement simulations were performed using mathematical transformations, while chest lead misplacement scenarios were simulated by interchanging pairs of leads. The detection performance was assessed using metrics such as accuracy, precision, sensitivity, specificity, and Macro F1-score. *Main results*. Our experiments simulated three scenarios of limb lead misplacement and nine scenarios of chest lead misplacement. The proposed two models achieved Macro F1-scores ranging from 93.42% to 99.61% on two heterogeneous test sets, demonstrating their effectiveness in accurately detecting lead misplacement across various arrhythmias. *Significance*. The significance of this study lies in providing a reliable open-source algorithm for lead misplacement detection in ECG recordings. The source code is available at https://github.com/wjcai/ECG_lead_check.

1. Introduction

The 12-lead electrocardiogram (ECG) is a widely used diagnostic tool for heart disease that records the electrical activity of the heart through electrodes placed on the body surface. The standard 12-lead ECG includes four limb leads (LA, RA, LL, and RL) and six chest leads (V1–V6) that provide crucial information for diagnosing various cardiac arrhythmias. However, the accurate placement of electrodes is essential for obtaining correct ECG recordings, as misplacement can result in incorrect diagnoses and treatment of heart disease. The incidence of lead misplacement ranged from 0.4% to 4% for various types of leads misplacement (Hede *et al* 1995, Kors and van Herpen 2001, Rudiger *et al* 2007, Nilsson *et al* 2008) with limb leads being more susceptible than chest leads (Riddle 2008). Moreover, RA-LA misplacement was the most common type, occurring in approximately 70% of all misplacement cases (Hede *et al* 1995).

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Although lead misplacement is a relatively low-probability event, it can significantly impact accurate ECG interpretation, causing misdiagnosis and inappropriate treatment. Lead misplacement can bring significant changes to ECG waveform morphology and cause pathological patterns like ectopic rhythm, conduction disturbance (Dhingra *et al* 1975, Castellanos *et al* 2002, Wang *et al* 2022), ventricular hypertrophy or ventricular pre-excitation (Green and Palma 2003), simulating or concealing of myocardial ischemia or infarction (Horwitz and Medrano 1976, Chanarin *et al* 1990, Rudiger *et al* 2003, Akel *et al* 2004, Marafioti and Variola 2004, Batchvarov *et al* 2007). Some pathological patterns resulting from lead misplacement are difficult to distinguish, such as the case in Wang *et al* (2022), LA-LL can produce the patterns of alternating bundle-branch block and is hard to distinguish. Moreover, with the large number of ECG records in clinical practice, it is necessary to develop an effective detection method to reduce the workload of clinicians. Currently, some ECG machines have built-in lead misplacement detection algorithms, such as Philips's ECG machines with DXL ECG Algorithm (PHILIPS 2022) and Schiller's CARDIOVIT AT-102 G2 (SCHILLER 2021). However, the algorithms integrated into ECG machines are typically proprietary and closed systems, limiting the ability to customize or improve their performance. This lack of transparency and accessibility prevented researchers and developers from making advancements in lead misplacement detection and hampers the overall progress in this field. With the rise of wearable devices, there was a growing trend of collecting ECG data through these devices for remote monitoring and automated analysis. However, many emerging wearable devices did not have built-in lead misplacement detection algorithms. Consequently, when these ECG recordings were uploaded to servers for automated analysis, it became crucial to have a reliable algorithm that can effectively analyze and detect lead misplacements.

Traditional rule-based methods, such as lead P-wave orientation (Surawicz and Knilans 2001), the presence of leads with flat signal amplitude (Hoffman 2007), and QRS waveform amplitude (Haisty *et al* (1993)), have been used for lead misplacement detection. However, these methods were inherently constrained by their dependency on specific features and their limited adaptability to different types of lead misplacement scenarios.

With the advancement of computer and machine learning technology, automatic detection algorithms have been developed that use machine learning models to extract and classify features. These algorithms have shown promising results in detecting lead misplacement, but they were not without their limitations. Morphological and statistical features, such as P-QRS-T amplitude (Hede *et al* 1995, de Bie *et al* 2014, Gregg *et al* 2017), QRS wave boundary and QRS-T wave pattern (Jekova *et al* 2013), and P-QRS-T vector direction (Han *et al* 2014), have been selected to identify lead misplacement. In Rjoob *et al* (2019a, 2019b), authors conducted V1 and V2 misplacement detection in three different positions: first, second and third intercostal spaces, which was based on morphological, frequency, and statistical features. Lead correlation features calculate correlation coefficients between leads and use machine learning models to detect misplacement based on coefficient trends (Jekova *et al* 2013, Jekova *et al* 2016, Jekova *et al* 2018). Lead redundancy information utilized the fact that limb leads can be reconstructed through any two limb leads, and changes in the correlation coefficients of the reconstructed leads can be used to detect misplacement (Kors and van Herpen 2001). Other methods used the relationship information among leads include work in Maan *et al* (2011), which compared the error between the original ECG and the reconstructed ECG produced by Kor's transform (Kors *et al* 1990), and work in Starc (2011), which was based on the determination of equivalent dipoles. However, these automatic detection methods based on extracted features were often susceptible to noise and lack generalization ability and robustness. Moreover, detecting LA-LL is the most challenging limb lead misplacement task (Rjoob *et al* 2020) and several previous studies excluded the LA-LL detection from the experiment due to the detection difficulty (Han *et al* 2014) and inapplicability of the method (Jekova *et al* 2016). The existing studies of LA-LL detection have not achieved satisfactory results, with sensitivity ranging from 17.9% to 84.5% and specificity ranging from 95.6% to 99.7%. (Hede *et al* 1995, Hedén *et al* 1996, Kors and van Herpen 2000, Kors and van Herpen 2001, de Bie *et al* 2014, Gregg and Babaeizadeh 2020). Although LA-LL is less common, it can produce patterns of alternating bundle-branch block, which may lead to the patient being fitted with an unnecessary pacemaker (Glikson *et al* 2022, Wang *et al* 2022).

Recent advances in deep learning models have shown great promise in end-to-end automatic lead misplacement detection. For instance, Gregg and Babaeizadeh (2020) utilized a CNN model that directly inputs ECG segments to detect LA-LL misplacement, resulting in a significant improvement in detection accuracy with a sensitivity of 84.5% and a specificity of 99.5%. Rjoob *et al* used a deep learning model to detect V1 and V2 misplacement and compared the performance with feature engineering-based method, showed the better performance of deep learning model (Rjoob *et al* 2021). However, the existing research using deep learning in this domain remains limited, particularly in the area of multi-class classification encompassing normal leads and multiple types of misplacements.

To address the aforementioned limitations, we conducted a comprehensive investigation focusing on multiple types of ECG leads misplacement. Furthermore, we have developed an open-source algorithm to tackle this problem. We evaluated our algorithm on two heterogeneous datasets that contained various arrhythmias,

demonstrating its strong generalization ability and robustness. With a small number of parameters, the proposed deep learning model could effectively enhance the reliability of lead misplacement detection.

2. Methods

2.1. Datasets

In our experiment, we used three 12-lead ECG databases: the Chapman database (Gregg and Babaeizadeh 2020, Zheng *et al* 2020, Zheng *et al* 2022), the Physikalisch-Technische Bundesanstalt (PTB) database (Bousseljot *et al* 1995), and the PTB-XL database (Wagner *et al* 2020). We used 80% of the Chapman database for training and 20% for validation. The PTB-XL database and the PTB database were used for testing. Besides, the Lobachevsky University Electrocardiography Database (LUDB) (Kalyakulina *et al* 2020, Kalyakulina *et al* 2021) was used to examine the interpretability of the proposed model; details can be found in the section 2.5. All databases are available on Physionet (Goldberger *et al* (2000)).

(1) Chapman database

Chapman database contains standard 12-lead ECG records from 45 152 subjects, with 56% males and 44% females. The highest age groups prevalence were 51–60, 61–70, and 71–80 years representing 19.8%, 24%, and 17.3% respectively. Records include 63 types of arrhythmias. Each record is 10 s long and sampled at 500 Hz.

(2) PTB-XL database

PTB-XL database contains 21 837 ECG records from 18 885 subjects, with 52% males and 48% females, aging from 0 to 95 years. The record length is 10 s, and data sample rates of 100 Hz and 500 Hz are provided. We used the 500 Hz data in our experiment. Records are classified into five categories: Normal ECG (NORM), Myocardial Infarction (MI), Conduction Disturbance (CD), Hypertrophy (HYP), and ST/T change (STTC). Pacemaker rhythms were removed.

(3) PTB database

PTB database contains 549 standard 12-lead ECG records from 290 subjects, 209 are males and 81 are females, aging from 17 to 87 years, with the record length ranging from 32 s to 120 s, which sampled at 1000 Hz. Each record has 15 channels. The first 12 channels are for standard 12-leads ECG, and the remaining channels are for Frank ECG. Among the 290 subjects, 52 are healthy controls, and the rest suffer from various arrhythmias. We removed four LA-RA ECG records ('s0197lre', 's0196lre', 's0236lre', 's0166_re') and an unacceptable quality ECG record ('s0544_re') as in Jekova *et al* (2016).

(4) LUDB database

LUDB database consists of 200 10 s 12 lead ECG records sampled at 500 Hz. The subjects include 115 males and 85 females, aging from 11 to 90 years. Each of the record contains the annotations of P, QRS, and T boundaries.

Data preprocessing consisted of three parts: denoising, downsampling, and normalization. Due to the change in signal length, the PTB database was first segmented into 10 s without overlap, and parts less than 10 s were discarded. We used a 5-order Butterworth bandpass filter to denoise the signals, with a lower limit of 0.5 Hz and an upper limit of 35 Hz. In order to reduce the time cost of model training and inference, we referred to the method in (Fan *et al* 2018, Zhang *et al* 2022) and downsampled the signals to 120 Hz using the python Scipy library, which is based on the Fourier transform. Finally, we used Z-Score to normalize the ECG signals to reduce the differences among patients and increase the model's generalization ability. Let x_i be the ith lead of a sample X and $X = \{x_1, \ x_2, \cdots\cdots x_m\}$, where m is the number of leads in a sample. The calculation formula is defined as follows:

$$\hat{x}_i = \frac{x_i - \overline{x}_i}{s_i} \quad , \tag{1}$$

where $\overline{x}_i$ is the average amplitude of x_i, s_i is the standard deviation of signal amplitude, $\hat{x}_i$ represents the normalized ith lead signal of X.

2.2. Simulation of leads misplacement

The ECG recording involves placing four limb lead electrodes on the left wrist, right wrist, left ankle, and right ankle, that placement of 4 leads can be performed in 24 different ways, but only 1 way is correct. So 23 ways are there to simulate the misplacement. However, as the potential difference between the left and right ankles is negligible, only 12 different arrangements need to be simulated (Xia *et al* 2012). In addition, we excluded the

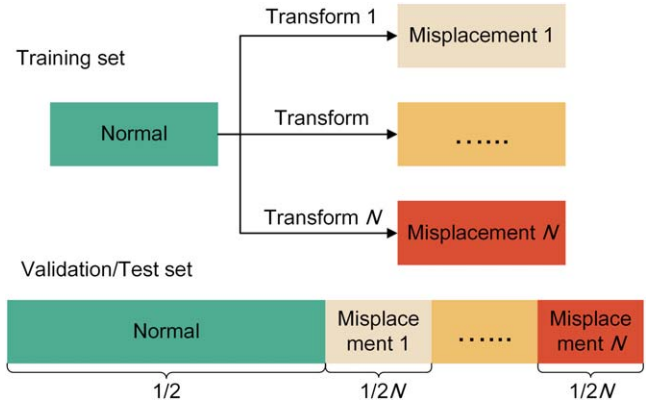

Figure 1. The process of creating datasets. *N* is the number of misplacement categories. For the training set, all misplacement categories have the same number as the Normal samples. For validation and test set, the total number of all the misplacement samples is the same as the number of Normal samples, each misplacement category has the same number of samples.

Table 1. Simulations of limb leads misplacement.

Class	Leads ordering						
Normal	I	II	III	aVR	aVL	aVF	V1–V6
LA-RA	−I	III	II	aVL	aVR	aVF	V1–V6
RA-LL	−III	−II	−I	aVF	aVL	aVR	V1–V6
LA-LL	II	I	−III	aVR	aVF	aVL	V1–V6

misplacement scenario where the right ankle electrode (RL) is mistakenly placed on the upper limb, as it can be easily detected due to the significant change in signal amplitude. Finally, according to the frequency statistics of lead misplacement in (Hede *et al* 1995), for limb lead misplacement, we simulated three different scenarios: LA-RA, RA-LL, and LA-LL reversal; simulation was done by mathematical transformation, as shown in table 1. For chest leads misplacement, we simulated all one-hop (V1–V2, V2–V3, …, V5–V6 reversal) and two-hop (V1–V3, V2–V4, …, V4–V6 reversal) scenarios, as these scenarios are prone to occur due to the relatively short distance. The misplacement of chest leads was simulated by exchanging two chest leads.

2.3. Methodology

To detect limb lead misplacement, we input two of the limb leads and V6 into the proposed lightweight deep learning model for automatic classification. The choice of the two limb leads was based on the model's best Macro F1-score on the validation set. The reason for using only two limb leads was that if two of the six limb leads are known, the remaining four limb leads can be calculated by a linear expression. In addition, only using two limb leads makes the inference faster and more efficient. Inspired by the work in Jekova *et al* (2016), we added V6 as the judgment basis, as V6 is the only chest lead located in the frontal plane, which can improve the detection effect. To detect chest lead misplacement, we used leads V1 to V6 as the input of the proposed model to obtain the classification result. For the training set, different simulations were performed on all samples in turn to obtain training set data with the same amount of each class; for the validation set and test set, 50% of the samples were randomly selected as normal samples, and different simulations were performed on the remaining 50%, with the same proportion of each class, which different misplacement scenarios would not appear for the same original ECG. The process of creating datasets is shown in figure 1.

2.4. Proposed models

For limb lead misplacement, we proposed a lightweight DenseNet model (LDenseNet) for the classification task. DenseNet (Huang *et al* 2017) was mainly composed of DenseBlock and TransitionBlock, addressing the problem of vanishing gradient in deep networks, reducing the number of model parameters, and had excellent feature extraction ability by using feature reuse. The DenseBlock contained Bottleneck and convolution layers, as shown in figure 2(b). Bottleneck were point-wise convolution layers, which could improve channel-wise communication, and each input of convolution layers came from all layers before. 1D-DenseNet has been applied in several ECG-related tasks (Xiong *et al* 2021, Bui and Pham 2023, Pradhan and Srivastava 2023). Since the characteristic of lead misplacement is on a lead scale, a complex deep model architecture is not necessary. To

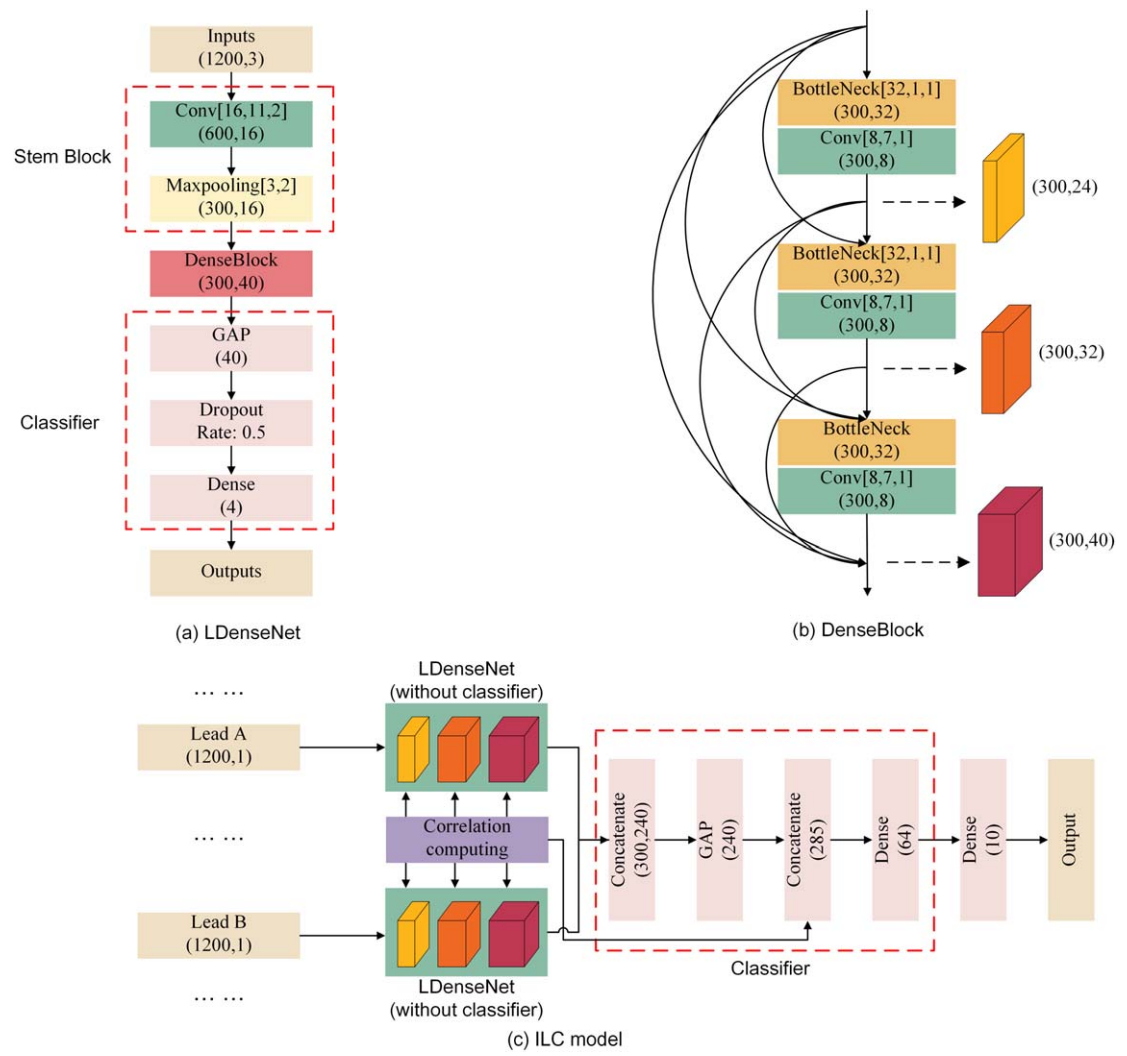

Figure 2. Model architecture. Conv represents convolution layer. [.] presents the parameters of layers. For the convolution layer, the order is kernel number, kernel size, and stride; for the MaxPooling layer, the order is kernel size and stride. (.) presents the output shape of layers.

improve the model inference efficiency, we simplified the model structure and parameter quantity, as was similar in Bui and Pham (2023). It was only contained one Dense Block. The structure is shown in figure 2(a). The input of LDenseNet was two limb leads with V6, and the choice of two leads depended on the models' best Macro F1-score on validation set. The feature of input ECGs was preliminarily extracted by Stem Block, which was composed of a Convolution layer and a MaxPooling layer. Then, the output of the Stem Block was sent to a Dense Block for further feature extraction, and the feature map output by the Dense Block was sent to the classifier for final prediction. The classifier consisted of a global pooling layer, a Dropout layer, and a fully connected layer. The output dimension was four, which corresponded to Normal (without any limb lead misplacement), LA-RA, RA-LL, and LA-LL, respectively. The model's output was activated by SoftMax, and the final prediction result was the one corresponding to the highest activation value.

For chest lead misplacement detection, due to the similarity between two adjacent leads, it is difficult to recognize the interchange between them. Inspired by Jekova *et al* (2016), which calculated the correlation coefficient between two chest leads to measure the distance between them, we proposed the Inter-lead correlation (ILC) model, as shown in figure 2(c). The model used a multi-branch structure and each branch was an LDenseNet. The input for each branch was a lead of a record. Therefore, the ILC model had six branches, and the input of each branch corresponded to V1–V6. After each branch extracted the feature map of the corresponding lead, all feature maps were concatenated in channel dimension and passed through a global average pooling layer. Then, the output feature maps of three convolutional layers of Dense Block in each branch were selected to calculate the correlation coefficients with each other. The number of different branch combinations was $C(6,2) = 15$, and for each combination, the correlation coefficient between three feature maps were calculated respectively (with the total correlation coefficient number of 45). The correlation coefficients

Table 2. Model parameters. n is the number of convolution kernels, N is the number of Convolution layers, ks is the size of the convolution kernel/pooling kernel, and s is the stride.

Block	Parameters	Search window	Optimal parameters
Stem block	Convolution layer	$n = 8, 16, 32,$ $ks = 7, 9, 11, 13, s = 2$	$n = 16, ks = 11, s = 2$
	Activation layer	Swish, ReLU, GELU	Swish
	Maxpooling	$ks = 3, s = 2$	$ks = 3, s = 2$
Dense block	Number of convolution layer	$N = 1, 2, 3, 4$	$N = 3$
	Bottleneck	$n = 16, 32, 64,$ $ks = 1, s = 1$	$n = 32, ks = 1, s = 1$
	Convolution layer	$n = 4, 8, 16,$ $ks = 5, 7, 9, 11, s = 1$	$n = 8, ks = 7, s = 1$
	Activation layer	Swish, ReLU, GELU	Swish
Classifier	Dropout	/	Rate = 0.5
	Dense	/	Output dims: 4 (LDenseNet) 64, 10 (ILC model)
	Activation layer	/	Hidden layer: Relu Output layer: Softmax

were combined with the branch feature maps and then passed through the classifier to get the final output. The output dimension was 10, which corresponded to Normal (without any chest lead misplacement) and different chest lead misplacement scenarios (V1–V2, …, V5–V6). Same as the limb lead misplacement detection, the model's output was also activated by SoftMax, and the category with the highest activation value was the final prediction result. Besides, the parameters of branches were shared, which could reduce the model's parameters and ensure the consistency of the distance between different leads, as different parameters in branches might disrupt the similarity relationship between leads. Mark X as samples, $X = \{x_1, x_2, x_3, x_4, x_5, x_6\}$, $L_0(\cdot)$ as the Stem Block, and the Layer of DenseBlock as $L_n(\cdot)$, where n is the layer index, $n\epsilon\{1,2,3\}$, thus the output feature maps of Stem Block and layers of DenseBlock can be defined as:

$$f_{L_0}^{(k)} = L_0(x_k) \tag{2}$$

$$f_{L_1}^{(k)} = L_1(f_{L_0}^{(k)}) \tag{3}$$

$$f_{L_2}^{(k)} = L_2(\mathrm{Concat}(f_{L_0}^{(k)}, f_{L_1}^{(k)})) \tag{4}$$

$$f_{L_3}^{(k)} = L_3(\mathrm{Concat}(f_{L_0}^{(k)}, f_{L_1}^{(k)}, f_{L_2}^{(k)})), \tag{5}$$

where k is the lead index, $k \in \{1, 2, 3, 4, 5, 6\}$, Concat refers to the concatenate function.

Then correlation coefficients between feature maps will be calculated. The correlation coefficient is defined as:

$$r(f^{(k)}, f^{(k+1)}) = \frac{\sum_{ij} f^{(k)} \cdot f^{(k+1)}}{\|f^{(k)}\|_2 \times \|f^{(k+1)}\|_2}, \tag{6}$$

where $f^{(k)}$ refers to feature map of lead k, $f^{(k)} \in R^{ij}$; dot refers to the element-wise product, and $\|\cdot\|_2$ refers to the 2-norm.

The correlation coefficient for each layer can be defined as follows:

$$\begin{aligned} &Cof_{L_n} \\ &= [r(f_{L_n}^{(k)}, f_{L_n}^{(k+1)}), r(f_{L_n}^{(k)}, f_{L_n}^{(k+2)}), \ldots, r(f_{L_n}^{(k)}, f_{L_n}^{(k+m)})], \\ &k\epsilon\{1, 2, 3, 4, 5, 6\}, k + m \leqslant 6, \quad n\epsilon\{1, 2, 3\}. \end{aligned} \tag{7}$$

The final prediction of the ILC model can be describe as:

$$\begin{aligned} &\mathrm{Prediction} = \mathrm{Classifier}(\mathrm{Concat}(f_{L_n}^{(k)}, Cof_{L_n})), \\ &k\epsilon\{1, 2, 3, 4, 5, 6\}, \quad n\epsilon\{1, 2, 3\}. \end{aligned} \tag{8}$$

The specific parameters of the model are shown in table 2, using the Bayesian Optimization algorithm implemented by Keras Tuner (O'Malley *et al* 2019), and the metric was the Macro F1-score on the validation set. The padding mode for all convolution layers was set to 'same', using swish as the activation function. The number of trainable parameters of LDenseNet and ILC model is 8508 and 26 946. A Dense block contains 7800 trainable parameters. We used categorical cross-entropy loss for model training, with Adam as the optimizer.

For limb lead misplacement, 0~4 presented Normal, LA-RA, RA-LL, and LA-LL, respectively. For chest lead misplacement, 0~9 presented Normal, V1–V2, V1–V3, V2–V3, V2–V4, V3–V4, V3-V5, V4–V5, V4–V6, V5–V6, respectively. The initial learning rate was set to 0.001, and the batch size was set to 512, with the learning rate decay of the initial learning rate/Epoch number. The experiments were performed on the Windows 10 system, using Tensorflow 2.4.0 as the deep learning framework and Nvidia GeForce RTX 3060 for training. CPU model was AMD Ryzen 5 5600X.

2.5. Model interpretability

Deep learning models are often limited in practical applications due to their lack of interpretability. To address this issue, many interpretability methods for deep learning models have been proposed. In this study, we used GradCAM ++ (Chattopadhay *et al* 2018) to investigate the interpretability of the model. GradCAM++ can reflect the impact of different parts of the input signal on the model's prediction and can be presented in as a heatmap. To quantify the model's interest in different waveforms, the LUDB database was chosen for experimentation, with each record containing the annotations of P wave, QRS complex, and T wave boundaries. Firstly, for a certain lead misplacement, simulation of this misplacement was performed on all samples in LUDB. Secondly, the model was used to predict the samples and get the activation maps, where the activation value ranged from 0 to 1. If the activation value was greater than 0.5, it was considered as a saliency region. We calculated the composition ratio of each waveform in the saliency regions. Since each different waveform has a different duration, different scaling is required for different waveform lengths when calculating the ratio. In detail, the scaling coefficients are calculated as follows:

$$\alpha = \left[\frac{\min\left(\mathrm{len}(A_P),\ \mathrm{len}(A_{\mathrm{QRS}}),\ \mathrm{len}(A_T),\ \mathrm{len}(A_{\mathrm{Others}})\right)}{\mathrm{len}(A_{\mathrm{wave}})}\right] \tag{9}$$

$$I = A \cap S \tag{10}$$

$$\mathrm{Rate} = \frac{\mathrm{len}(I_{\mathrm{wave}})\cdot\alpha_{\mathrm{wave}}}{(\mathrm{len}(I_P),\ \mathrm{len}(I_{\mathrm{QRS}}),\ \mathrm{len}(I_T),\ \mathrm{len}(I_{\mathrm{others}}))\cdot\alpha^T},$$

$$\mathrm{wave} \in \{P,\ QRS,\ T,\ \mathrm{Others}\}, \tag{11}$$

where A represents the annotation region of the waveform, S represents the saliency region, subscripts indicate different waveforms, and $len(\cdot)$ denotes the total number of the sampling points of all samples.

2.6. Metrics

In this study, we evaluated the model using Accuracy (Acc), Sensitivity (Se), Specificity (Sp) and Macro F1-score (Macro F1). The calculation formulas are shown as follows:

$$\mathrm{Acc} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}} \tag{12}$$

$$\mathrm{Prec} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{13}$$

$$\mathrm{Se} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{14}$$

$$\mathrm{Sp} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}} \tag{15}$$

$$\mathrm{F1} = \frac{2\mathrm{TP}}{2\mathrm{TP} + \mathrm{FN} + \mathrm{FP}} \tag{16}$$

$$\mathrm{Macro}\ F1 = \frac{F_{1A} + F_{1B} + \ldots + F1_N}{N}, \tag{17}$$

where TP is true positive, TN is true negative, FP is false positive, FN is false negative, F_{1A} presents the $F1$-score of class A, and N presents the number of samples.

To measure the computational efficiency of the model and compare it with methods based on machine learning models, we defined SPS (samples per second) metric to quantify the inference speed of the model, and the calculation is shown as follows: SPS = Total Samples Number/Total Time Cost. We used PTB-XL database to test all methods on CPU.

3. Results

3.1. Limb leads misplacement detection

By comparing the Macro F1-scores on the validation set, we selected leads III, aVR, and V6 as inputs for the model, as they achieved the highest Macro F1-score among the combinations of other limb leads and lead V6. The results on the test set for limb lead misplacement detection are shown in table 3, and the confusion matrixes are shown in figure 3. LDenseNet had excellent performance for detecting LA-RA and RA-LL, with the F1 scores around 99% on all three test sets. It also had a good performance in detecting LA-LL on these three test sets, with the sensitivity ranging from 93.9% to 97.0%, specificity from 98.5% to 99.4%, and F1 score from 93.5% to 97.0%.

The model achieved an overall accuracy of 97.7% in the PTB-XL database, 97.4% in the PTB database, and 98.5% in the LUDB. The F1 score of all classes in both test sets exceeded 93.4%. The confusion matrix showed that the model primarily encountered difficulties distinguishing between the normal and LA-LL reversal. This challenge arises from the fact that LA-LL reversal can lead to lead interchanges (e.g. leads I and II, aVL and aVF) and lead III inversion, thus making it intricate to accurately determine lead misplacement based solely on the polarity of P, QRS, and T waves.

3.2. Chest leads misplacement detection

The results for chest lead misplacement detection are shown in table 4, with a corresponding confusion matrix presented in figure 4. Notably, the F1-score for all chest lead misplacement scenarios exceeded 94.0% across both test sets. Among the various chest lead misplacement scenarios, V2–V3 reversal and V5–V6 reversal exhibited relatively lower sensitivity. This is attributed to the fact that these are adjacent leads with comparable waveforms and amplitudes, posing a challenge in accurately detecting these specific misplacement scenarios.

3.3. Ablation study

To further evaluate the effectiveness of our proposed model and method, we conducted ablation experiments. For limb lead misplacement detection, we compared the performance between using all limb leads and only using only a subset of two leads. Additionally, we experimented with excluding or including the V6 lead as input to investigate its impact. The results are summarized in table 5. It can be seen that using only leads III, aVR, and V6 as inputs achieved effects comparable to all limb leads combined with lead V6. This suggests that our approach requires less information and is more efficient than utilizing all limb leads. The inclusion of lead V6 resulted in a noticeable improvement in classification performance. On the PTB-XL database, the average precision, sensitivity, specificity, accuracy, and Macro F1-scores increased by 1.6%, 1.1%, 0.4%, 1.2%, and 1.4%, respectively. Similarly, on the PTB database, the improvements were 2.3%, 1.8%, 0.5%, 1.7%, and 2.1%, respectively. These results demonstrate the value of lead V6 as an input for detecting limb lead misplacement.

For chest lead misplacement detection, we analysed the performance differences between the LDenseNet and ILC model, along with the effects of weight sharing. Additionally, the contribution of the correlation coefficient computation was also examined by comparing different variables used for computing correlations. The results are shown in table 6. The symbols of the last three lines are consistent with section 2.4.

The proposed ILC model exhibited limited improvement over the LDenseNet model when the branch model weights were not shared. However, upon implementing weight sharing, the model's performance underwent a notable enhancement, resulting in a 1.3% increase in the Macro F1-score. This significant improvement underscores the rationality of the weight-sharing design. Furthermore, the inclusion of feature map correlation coefficient computing significantly improved the model performance on both two test sets. Specifically, the Macro F1-scores improved by a range of 3.5%–3.9% on the PTB-XL dataset and 3.6%–4.0% on the PTB dataset. Notably, the performance gains were positively correlated with the number of feature maps included in the computation, indicating that the increased feature map information contributed to the improved performance.

3.4. Benchmarks

We compared our approach with two established methods: one based on redundancy information from Kors and van Herpen (2001) and another leveraging correlation coefficient, as described by Jekova *et al* (2016). We refrained from including the morphological method in our comparison, as its feature extraction process is both cumbersome and susceptible to noise interference. The scenarios of limb lead misplacement conducted by Jekova *et al* (2016) and Kors and van Herpen (2001) were LA-RA, RA-LL, CW (clockwise rotation), CCW (counter-clockwise rotation), and Normal. LA-LL was not included, as these two methods include previous other studies cannot effectively identify LA-LL. The scenarios of chest lead misplacement were in line with this paper. Since these methods only focus on the binary classification problem between normal and a specific

Table 3. Results for limb lead misplacement detection.

Dataset	PTB-XL Acc: 97.7%				PTB Acc: 97.4%				LUDB Acc: 98.5%			
Class	Prec (%)	Se (%)	Sp (%)	F1 (%)	Prec (%)	Se (%)	Sp (%)	F1 (%)	Prec (%)	Se (%)	Sp (%)	F1 (%)
Normal	98.0	97.6	98.0	97.8	98.1	97.2	98.1	97.6	98.0	100.0	98.0	98.5
LA-RA	99.8	99.5	100.0	99.6	98.4	99.5	99.7	99.0	100.0	97.0	100.0	98.5
RA-LL	99.5	99.6	100.0	99.6	99.0	99.5	99.8	99.2	100.0	100.0	100.0	100.0
LA-LL	92.8	94.3	98.5	93.5	93.0	93.9	98.6	93.4	97.0	97.0	99.4	97.0
Avg ± std	97.5 ±2.8	97.7 ±2.2	99.1 ±0.8	97.6 ±2.5	97.1 ±2.4	97.5 ±2.3	99.0 ±0.7	97.3 ±2.3	98.8 ±1.3	98.2 ±1.3	99.3 ±0.8	98.5 ±1.1

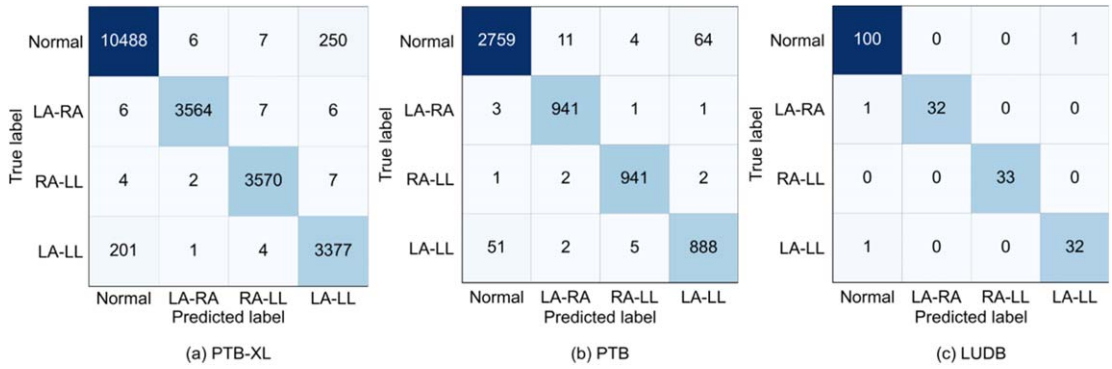

Figure 3. The confusion matrixes of LDenseNet on PTB-XL, PTB, and LUDB.

misplacement scenario (Normal versus LA-RA, Normal versus RA-LL, …), model ensemble was needed to carry out the binary classification problem between normal and multiple misplacement scenarios to meet the needs of practical application. We adopted a decision criterion where a positive misplacement indication from any model was considered a positive result, while a lack of such indication was regarded as negative. Similarly, for our multi-classification model, any abnormal output was deemed a positive misplacement, and vice versa.

The experimental results are shown in table 7. Models' details are shown in table 8. The inference process was all done on the CPU. It can be observed that the method proposed in this paper outperformed both the correlation coefficient-based method and the redundancy information-based approach. For limb lead misplacement detection, the Macro F1-score of our method was 2.0% and 4.5% higher than the redundancy information-based method on the PTB-XL and PTB database, respectively, and the correlation coefficient-based method had comparable performance to our proposed method. While the correlation coefficient-based method exhibited comparable performance to our proposed method, few previous studies have achieved satisfactory results on LA-LL reversal. The Se of these methods ranged from 17.90% to 84.50%, while the Sp fell between 95.60% and 99.97% (Hede *et al* 1995, Hedén *et al* 1996, Kors and van Herpen 2000, de Bie *et al* 2014). In contrast, our approach achieved a Se of over 93.9% and an Sp of over 98.6% on both test sets, as detailed in section 3.1. For chest lead misplacement detection, our proposed method also demonstrated significant performance improvements compared to both the correlation coefficient-based and redundancy information-based methods. Specifically, we observed an increase in Macro F1-score of 3.3% and 2.4% on the PTB-XL dataset, and 3.4% and 2.3% on the PTB dataset, respectively.

Regarding inference efficiency, the correlation coefficient-based method exhibited a clear advantage due to its straightforward feature extraction process and the simplicity of the required classifiers. In contrast, the redundancy information-based method involved a more intricate feature extraction process, leading to a relatively higher time cost. The method introduced in this paper maintained a high level of inference efficiency, attributed to its minimal parameter count. However, it is worth noting that in chest lead misplacement detection, the inference speed decreased slightly due to the additional parameters introduced by the Dense layers and the computation of correlation coefficients.

3.5. Model interpretation

Figure 5 illustrates the model heatmaps for limb lead misplacement scenarios. Figure 6 displays the heatmap of a V1–V2 misplacement scenario. The saliency region's composition reflects the primary criteria used by the model to distinguish between different misplacement scenarios. For LA-RA reversal, the LDenseNet mainly focused on the QRS wave. This can be attributed to the typical downward QRS wave in lead aVR, while the QRS waveform in lead aVL tends to be more variable. For RA-LL reversal, the LDenseNet mainly focused on the P wave and QRS wave, as the QRS waveform in lead aVF was mainly upward and the P wave in lead aVR was typically downward. For LA-LL reversal, the LDenseNet's attention was primarily drawn to the P wave. However, due to the variability of the P wave in lead III, it is challenging to provide a straightforward explanation. Regarding chest lead misplacement, the analysis clearly demonstrated that the ILC model primarily focused on the QRS wave as the basis for judgment. As the ratio of R wave amplitude to S wave amplitude gradually increases from V1 to V5, it becomes evident that the QRS region plays a crucial role in chest lead misplacement detection. The quantified results are presented in figure 7. It shows the proportion of each ECG waveform in the models' saliency region to different lead misplacement situations, which is consistent with the heatmap distribution in figures 5 and 6.

Table 4. Results for chest lead misplacement detection.

Dataset	PTB-XL Acc: 97.8%				PTB Acc: 98.3%				LUDB Acc: 99.0%			
Class	Prec (%)	Se (%)	Sp (%)	F1 (%)	Prec (%)	Se (%)	Sp (%)	F1 (%)	Prec (%)	Se (%)	Sp (%)	F1 (%)
Normal	99.5	96.5	99.5	98.0	99.9	97.6	99.9	98.7	100.0	98.0	100.0	99.0
V1–V2	97.8	99.3	99.9	98.6	99.1	99.7	99.9	99.4	100.0	100.0	100.0	100.0
V1–V3	99.3	99.9	100.0	99.6	99.4	99.7	100.0	99.5	100.0	100.0	100.0	100.0
V2–V3	91.1	98.2	99.4	94.6	93.2	99.4	99.6	96.2	100.0	100.0	100.0	100.0
V2–V4	98.3	99.5	99.9	98.9	98.8	100.0	99.9	99.4	100.0	100.0	100.0	100.0
V3–V4	98.0	99.0	99.9	98.5	97.8	97.8	99.9	97.8	91.7	100.0	99.5	95.7
V3–V5	99.2	99.9	100.0	99.5	99.1	99.1	99.9	99.1	100.0	100.0	100.0	100.0
V4–V5	96.6	99.2	99.8	97.9	93.7	98.7	99.6	96.1	91.7	100.0	99.5	95.7
V4–V6	96.5	99.3	99.8	97.9	96.3	98.4	99.8	97.3	100.0	100.0	100.0	100.0
V5–V6	90.3	97.9	99.3	94.0	94.5	98.7	99.7	96.6	100.0	100.0	100.0	100.0
Avg ± std	96.7 ±3.1	98.9 ±0.1	99.8 ±0.2	97.7 ±1.8	97.2 ±2.4	98.9 ±0.8	99.8 ±0.1	98.0 ±1.3	98.3 ±3.3	99.8 ±0.6	99.9 ±0.2	99.0 ±1.7

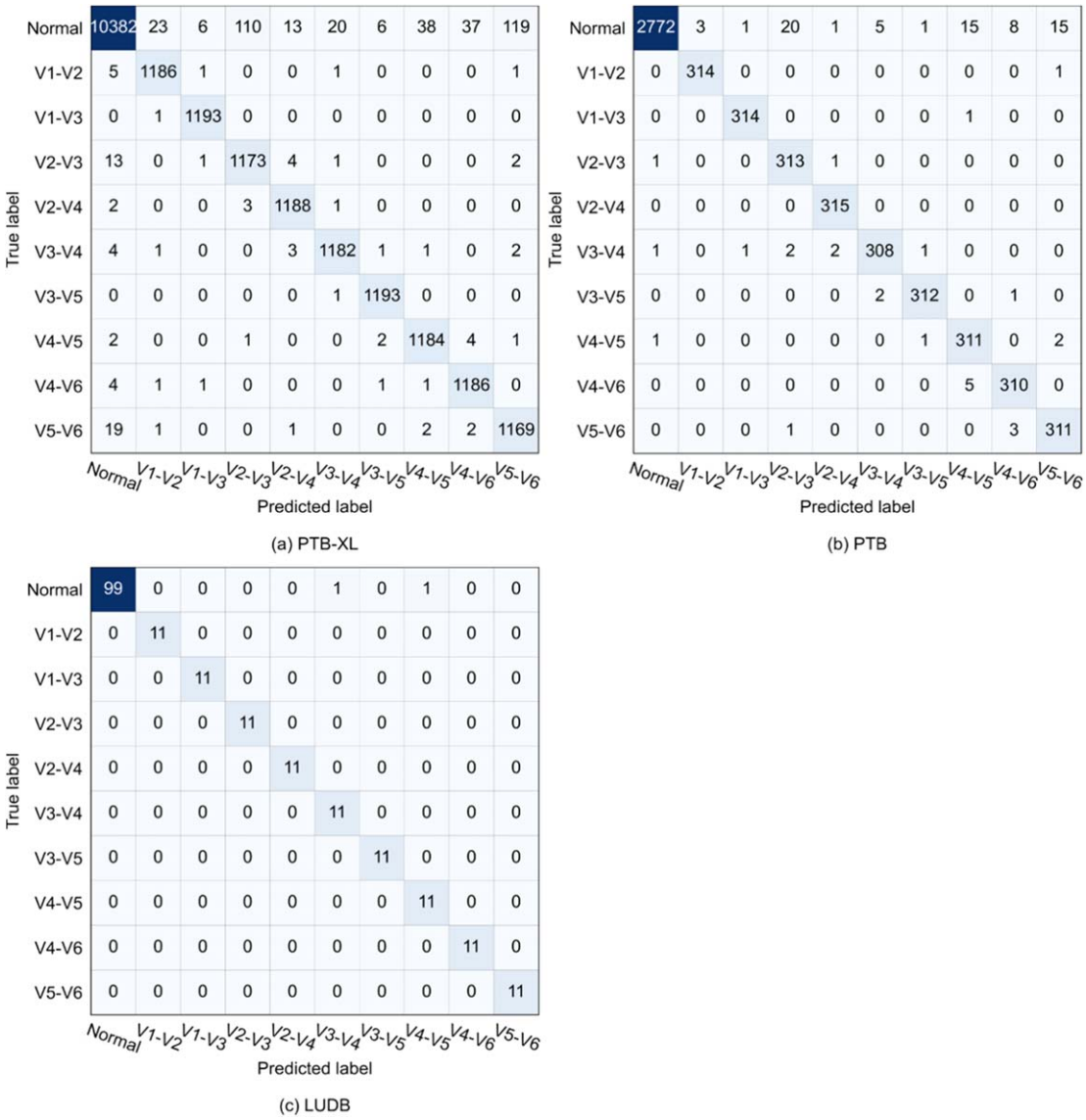

Figure 4. The confusion matrixes of ILC model on PTB-XL, PTB, and LUDB.

Table 5. Model performance with different inputs. 'A' represents leads III and aVR, 'B' represents all limb leads, and 'C' represents leads III, aVR, and calculated leads I, II, aVL, and aVF. All metrics are the average result. V6 improved the performance, and the performance of A+V6 input is comparable to that of B+V6, with less data volume.

	PTB-XL				PTB			
Input	Prec	Se	Sp	Macro F1	Prec	Se	Sp	Macro F1
A	95.9	96.6	98.7	96.2	94.8	95.7	98.5	95.2
B	96.7	97.1	98.9	96.9	95.5	96.4	98.7	96.0
C	95.5	96.8	98.7	96.1	95.3	96.6	98.8	95.9
A+V6	**97.5**	97.7	**99.1**	**97.6**	97.1	97.5	99.0	97.3
B+V6	97.2	**97.8**	99.1	97.5	**97.7**	**97.9**	**99.2**	**97.8**
C+V6	97.0	97.4	99.0	97.2	97.1	97.2	98.9	97.1

4. Discussion

In this study, we proposed a lightweight deep learning model for limb lead misplacement detection and chest lead misplacement detection, respectively. For limb lead misplacement detection, we proposed LDenseNet, which contained only one Dense Block and only used III, aVR, and V6 as the model's input. For chest lead

Table 6. Macro F1-score of Different models on test set. Weight sharing and correlation computing were both helpful to improve the performance of the model. With more feature maps involved in correlation computing, model performance becomes better.

Model	PTB-XL	PTB
LDenseNet	96.7	97.0
ILC model (w/o weight sharing)	96.9	96.6
ILC model (w/o correlation computing)	94.1	93.7
ILC model (L_3)	97.6	97.3
ILC model (L_2, L_3)	97.6	97.7
ILC model (L_1, L_2, L_3)	**98.0**	**97.7**

Table 7. Different method performance of limb misplacement detection. The best value for each term is written in bold. The proposed method performs better than the other two machine learning-based methods in limb lead misplacement detection and chest lead misplacement detection.

Method	PTB-XL				PTB			
	Prec	Se	Sp	Macro F1	Prec	Se	Sp	Macro F1
Kors-limb	99.3	99.6	99.3	99.5	98.4	99.2	98.4	98.8
Jekova-limb	97.3	98.3	97.3	97.8	94.1	95.8	94.0	95.0
Proposed-limb	**99.7**	**99.8**	**99.7**	**99.8**	**99.3**	**99.7**	**99.3**	**99.5**
Kors-chest	91.7	98.2	91.1	94.8	92.7	98.2	92.3	95.4
Jekova-chest	93.6	97.8	93.3	95.7	95.0	97.9	94.9	96.4
Proposed-chest	**96.6**	**99.5**	**96.5**	**98.1**	**97.6**	**99.9**	**97.6**	**98.7**

Table 8. Model details and inference speed. The higher the SPS, the higher the inference efficiency of the method. The highest SPS for limb and chest lead misplacement detection is written in bold.

Method	Model	SPS
Kors-limb	Random Forest, tree num: 2	**8179**
Jekova-limb	Random Forest, tree num: 2	160
Proposed-limb	LDenseNet, parameters: 8508	5064
Kors-chest	Random Forest, tree num: 4	**4273**
Jekova-chest	Random Forest, tree num: 2	74
Proposed-chest	ILC model, parameters: 26 946	1069

misplacement detection, we proposed the ILC model, which was a multi-branch model based on LDenseNet, and introduced the inter-lead correlation coefficient computation into the model's forward propagation. We tested the models on three independent test sets (PTB-XL, PTB, and LUDB), which all contain various cardiac arrhythmias, and achieved satisfactory performance, showing good generalization of the models.

In limb lead misplacement detection, the model's performance was enhanced by incorporating V6. This improvement can be attributed to the fact that V6 is primarily located in the frontal plane, similar to lead I, and acts as an anchor lead. When comparing the use of only III and aVR as input versus using all limb leads as input, the results indicate that the computed leads did not contribute to improving the detection performance. One possible reason is that the remaining leads are determined through linear transformations of III and aVR, whereas the deep learning model itself already encompasses numerous linear and nonlinear transformations. Hence, precomputing additional leads did not offer significant benefits in enhancing the model's performance. Therefore, utilizing III, aVR, and V6 as the model's input yielded comparable results to using all limb leads and V6, while requiring less time for training and prediction.

In chest lead misplacement detection, an evident observation is that the detection performance decreases when two interchanged leads are closer to each other. Almost all metrics for V1–V2, V2–V3, V3–V4, and V4–V5 detection were lower than those for V1–V3, V2–V4, V3–V5, and V4–V6 detection, respectively. This phenomenon can be explained by the fact that as the amplitude of the R wave gradually increases from V1 to V5 leads, the waveform changes between two adjacent leads become smaller compared to the changes between two alternate leads. Furthermore, ECGs with poor R-wave progression poses a challenge to chest misplacement detection.

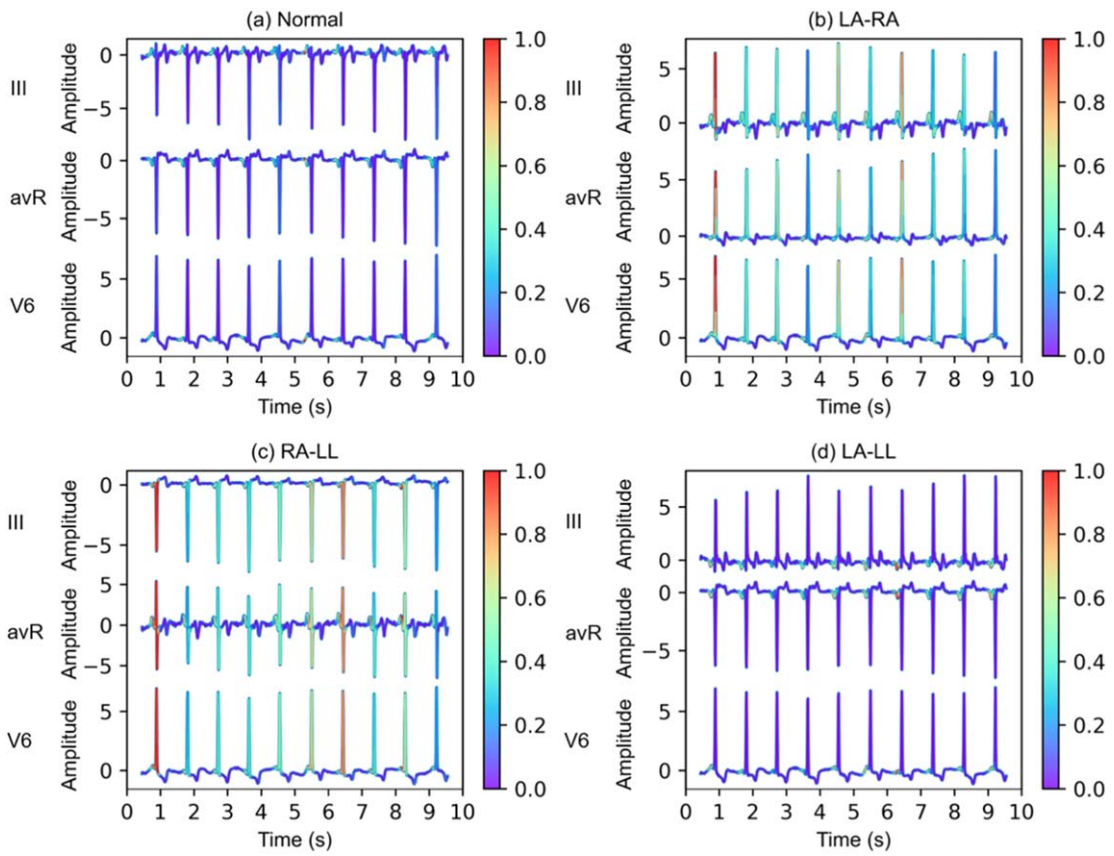

Figure 5. The GradCAM++-based heatmaps of LDenseNet for different limb lead misplacement samples. (a) Normal, model focused on the P wave, (b) LA-RA, on the QRS wave, (c) RA-LL, on the P wave and the QRS wave, and (d) LA-LL, on the P wave.

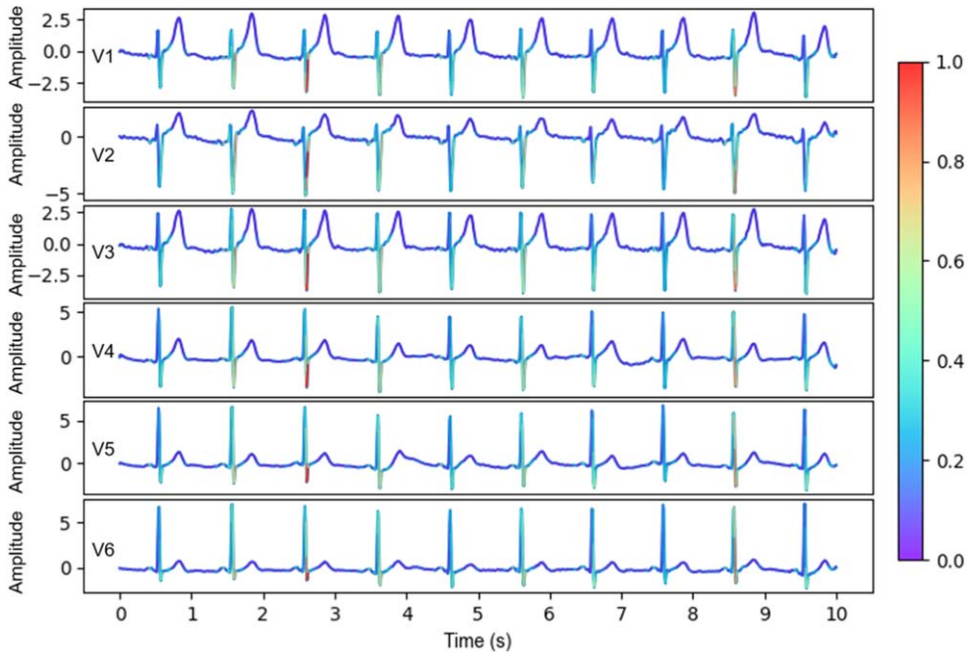

Figure 6. The GradCAM++-based heatmaps of the ILC model for a V1–V2 sample. The model focused on the QRS wave.

In practical applications, lead misplacement is a low-probability event, with the number of correctly recorded 12-lead ECGs far exceeding the number of lead misplacements. Therefore, the detection model not only needs high sensitivity but also high specificity to prevent frequent false positives from deterring users. For

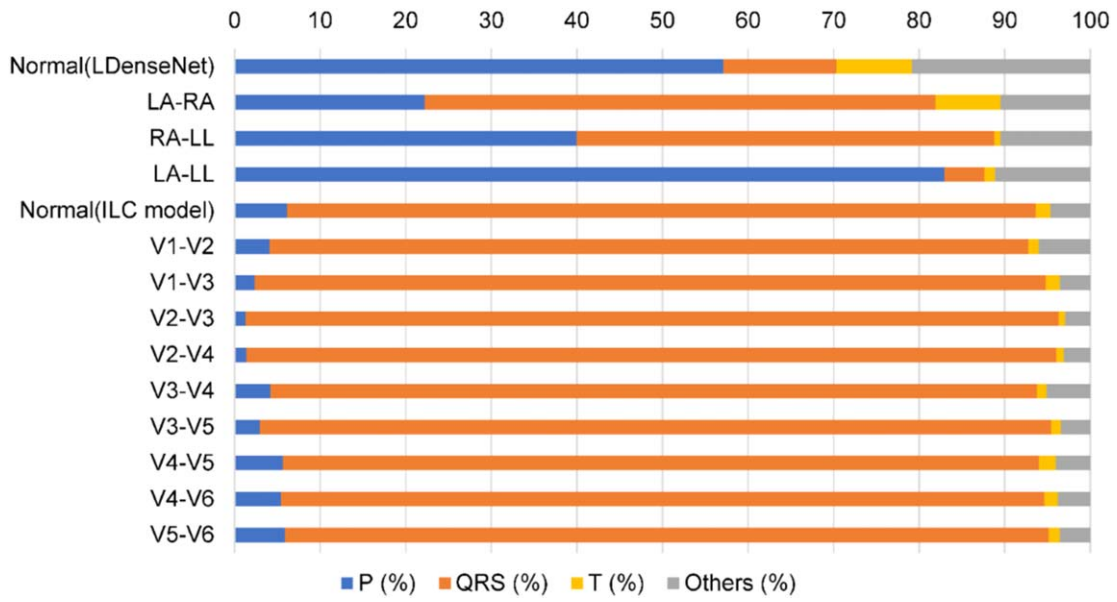

Figure 7. Proportion of each ECG waveform in models' saliency region to different lead misplacement. For Normal and LA-LL, model mainly focused on the P wave; For LA-RA, the QRS wave was the area of attention for the model; for RA-LL, model paid comparable attention to P wave and QRS wave; For all chest lead misplacement, QRS wave was the important region for model's judgement.

limb lead misplacement detection, the model specificity was 98.5% for LA-LL detection and over 99.2% for LA-RA and RA-LL. For chest lead misplacement detection, all scenarios had a specificity above 99.3%, demonstrating that the method satisfies this requirement. Additionally, ECGs collected from patients contain various arrhythmia and abnormal waveform patterns, demanding the model to detect lead misplacement not only for normal sinus rhythm but also in the presence of ECG abnormalities. Our models were well-suited to this demand, as we trained and validated the model using a large ECG database containing various arrhythmias and tested it on three external ECG databases that also have arrhythmias, achieving satisfactory results. Another consideration is the memory and computing power of hardware devices. Besides ECG machines and remote servers, our models are also suitable for deployment on resource-limited wearable devices. As lightweight models, LDenseNet has only 8508 parameters, and ILC has 26 946 parameters, with storage sizes of 0.11 MB and 0.33 MB, respectively. Another scenario where the models have high application value is telemedicine. For hospitals in remote areas, due to the lack of experienced experts, ECGs cannot be accurately interpreted, and it is often necessary to transmit the ECG data to an ECG remote diagnosis centre for experienced experts to interpret and return the diagnosis results for patient treatment. Because some lead misplacement scenarios are very concealed, such as the case of false alternating bundle-branch block caused by LA-LL mentioned in Wang *et al* (2022), this poses a great challenge to the knowledge level of experts. Lead misplacement detection can effectively improve the signal quality, help experts make accurate diagnoses, and ensure that patients receive reasonable treatment. Furthermore, a lightweight lead misplacement detection model can detect the collected ECG signals in real-time. The advantage is that misplacement can be found before the end of the collection and corrected in time, saving the time of recollection and improving the efficiency of examination.

To better evaluate the model's lead misplacement detection performance, we conducted predictions on a real clinical ECG dataset derived from the Remote ECG Diagnosis Centre of Shanghai Jiao Tong University School of Medicine, supported by Siwei Medical Co. Ltd., Shanghai, China. The dataset consisted of 1000 12-lead ECG records, which were 1000 consecutive cases received on a randomly selected day in 2023. Each record had a sampling rate of 500 Hz and a duration ranging from 10 to 60 s, encompassing a variety of arrhythmias. Two records were noted as lead misplacement samples: one involving LA-RA reversal and the other involving V3–V4 reversal. We selected the first 10 s of each record for lead misplacement detection. Our model successfully detected the two lead misplacement records, but it predicted many more positive samples. Specifically, in limb lead misplacement detection, the model predicted positive results in 34 records, while in chest lead misplacement detection, it predicted positive results in 58 records. Upon manual review by experts, it was discovered that clinicians had overlooked nine LA-RA, three V1–V2, two V2–V3, one V4–V5, and four V5–V6 misplacement records. It underscores the fact that clinicians often overlook numerous lead misplacement records in clinical practice. The significance of our algorithm lies in its ability to identify potential lead misplacements and subsequently enhance the quality of ECG signals. On the other hand, we found that a considerable part of the false positives was due to poor signal quality, including many unacceptable records, as

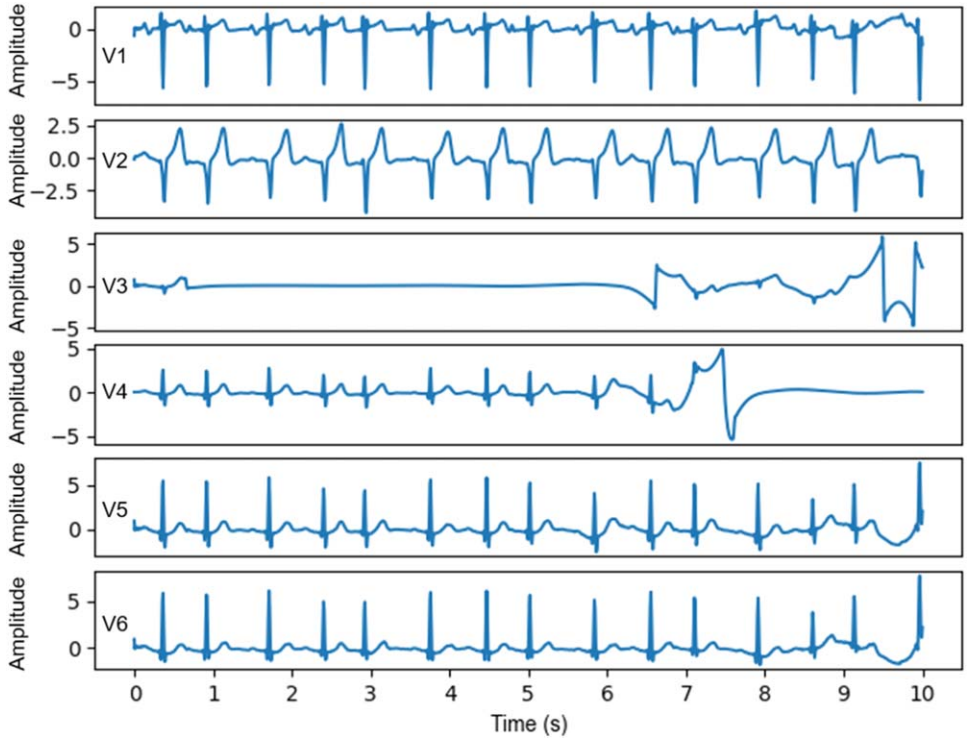

Figure 8. A V2–V3 sample predicted by ILC model. It is unjudgable since the unacceptable quality of V3.

shown in figure 8. For limb lead misplacement detection, 18 of the remaining 23 false positive records had poor signal quality, while for chest lead misplacement detection, it was 28 of the remaining 45 false positive records. To reduce the number of false positives, one approach is to post-process the model's output by artificially setting a threshold, and only when the confidence of the positive category is greater than the set threshold will the sample be classified as positive. We set thresholds of 0.9 and 0.7 for all classes of limb lead misplacement and chest lead misplacement, respectively. For limb lead misplacement detection, 16 false positives records were reduced without affecting the detection of true positive samples. For chest lead misplacement detection, it caused the missing detection of two true positive samples but reduced 21 false positive samples. Setting a threshold value on the model output has a minimal impact on true positive detection while significantly reducing the number of false positives.

While our study demonstrates the effectiveness of the proposed method in detecting limb and chest lead misplacement, there are several limitations that should be acknowledged. Firstly, the accuracy of the model relies on the correct placement of V6 as an input, which could be a limitation in certain scenarios. To address this, future studies could explore the use of redundancy information among limb leads to reduce the dependency on V6 when detecting limb lead misplacement. Secondly, the multi-branch structure and feature map-wise correlation coefficient calculation in the ILC model increase the time complexity, resulting in a slower inference speed than the LDenseNet. Therefore, there is room for improvement in terms of computational efficiency. Furthermore, when testing with real misplacement samples, we found that the model tended to report false positives when encountering records of poor signal quality. To mitigate this, we set decision thresholds on the model output, which reduced the number of false positives. Another potential approach is to combine an ECG signal quality evaluation algorithm to ensure ECG quality before performing misplacement detection.

5. Conclusion

The present study proposes a novel method for detecting limb and chest lead misplacement in ECG signals, which demonstrates superior performance compared to traditional methods. Our method utilizes a multi-classification model that can effectively differentiate on heterogeneous data sets containing various arrhythmias, showcasing its generalization and robustness. Additionally, the LDenseNet and ILC models employed in the study have a small number of parameters, which reduces the requirement for running memory and inference time, making them suitable for deployment on wearable devices. The findings of this study have the potential to significantly improve the accuracy of ECG recordings and reduce the misdiagnosis rate, ultimately leading to

better patient outcomes. Future studies can build upon these results by exploring methods to further enhance the efficiency and accuracy of the proposed method.

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Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: https://doi.org/10.13026/wgex-er52, https://doi.org/10.13026/x4td-x982, https://doi.org/10.13026/C28C71 and https://doi.org/10.13026/eegm-h675.

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